Longest path

Classic 1-D compaction sets each variable to the longest path from the source

The idea

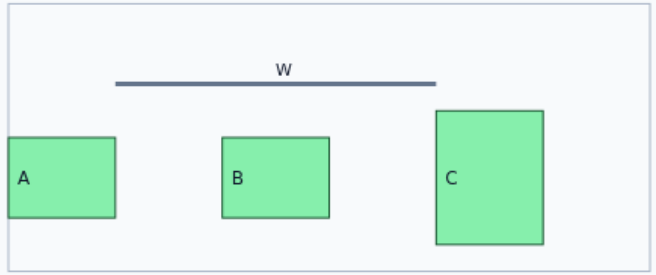
- Relax edges until positions stabilize; left edges pull rightward to minimum legal places

Source

LONGEST PATH

Source

Step 1 / 5 · src · module02-03-longest-path



Explanation

SRC_X at position 0.

- origin

```
0
```

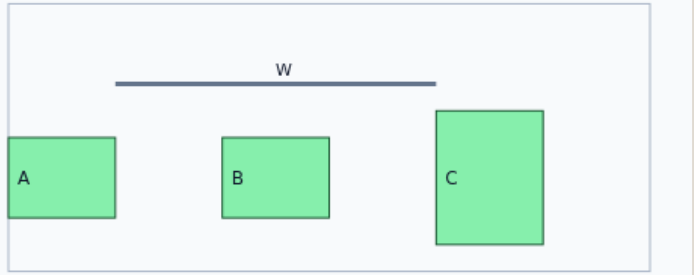
Source

Relax

LONGEST PATH

Relax

Step 2 / 5 · relax · module02-03-longest-path



Explanation

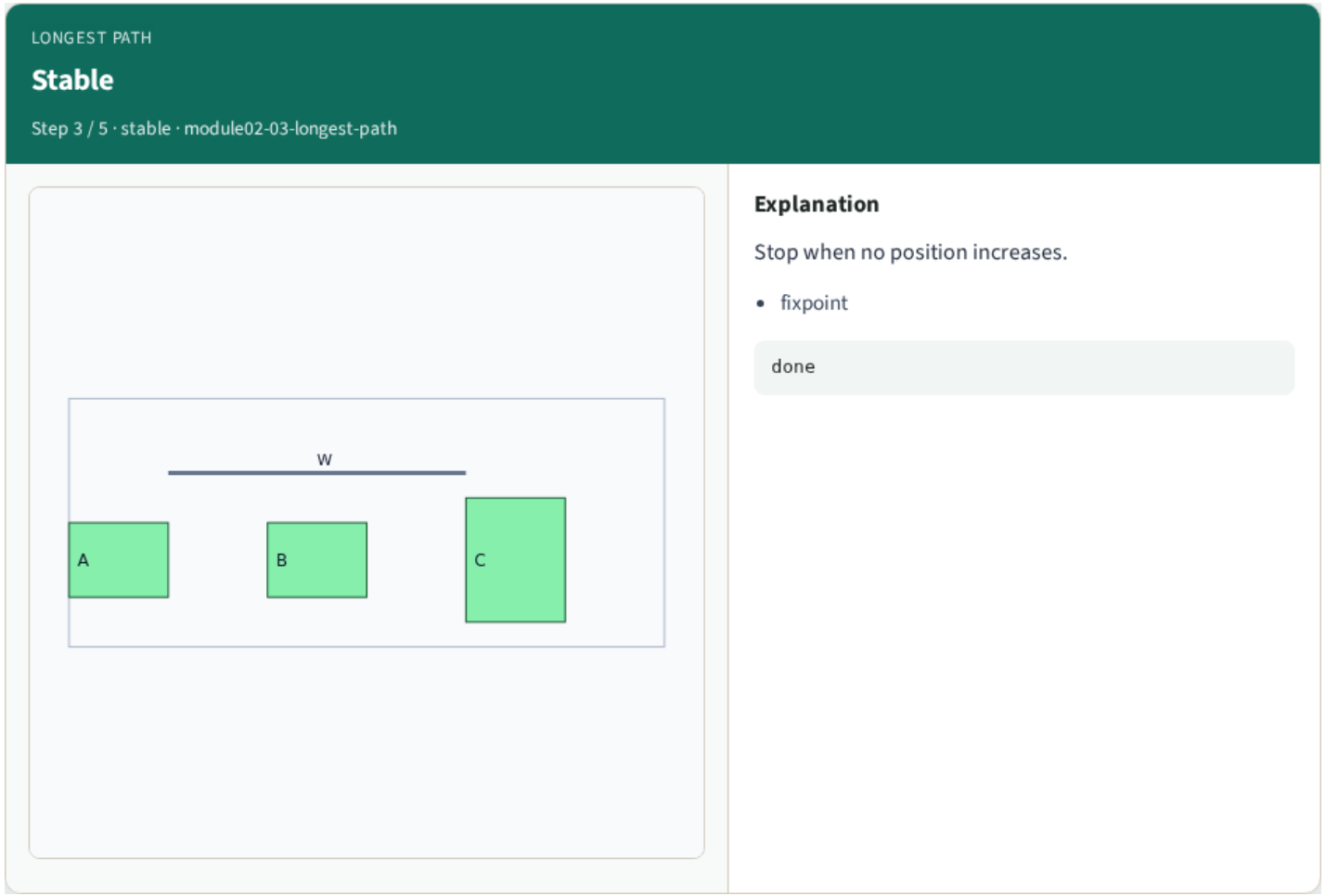
Update dst ← max(dst, src + w).

- Bellman-ish

```
relax
```

Relax

Stable



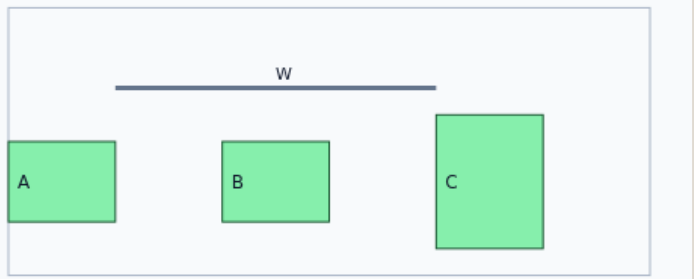
Stable

Left pack

LONGEST PATH

Left pack

Step 4 / 5 · pack · module02-03-longest-path



Explanation

Cells pack left under spacing.

- compact

```
toward SRC
```

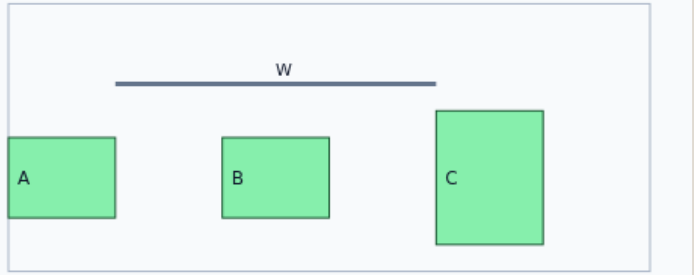
Left pack

Ready

LONGEST PATH

Ready

Step 5 / 5 · ready · module02-03-longest-path



Explanation

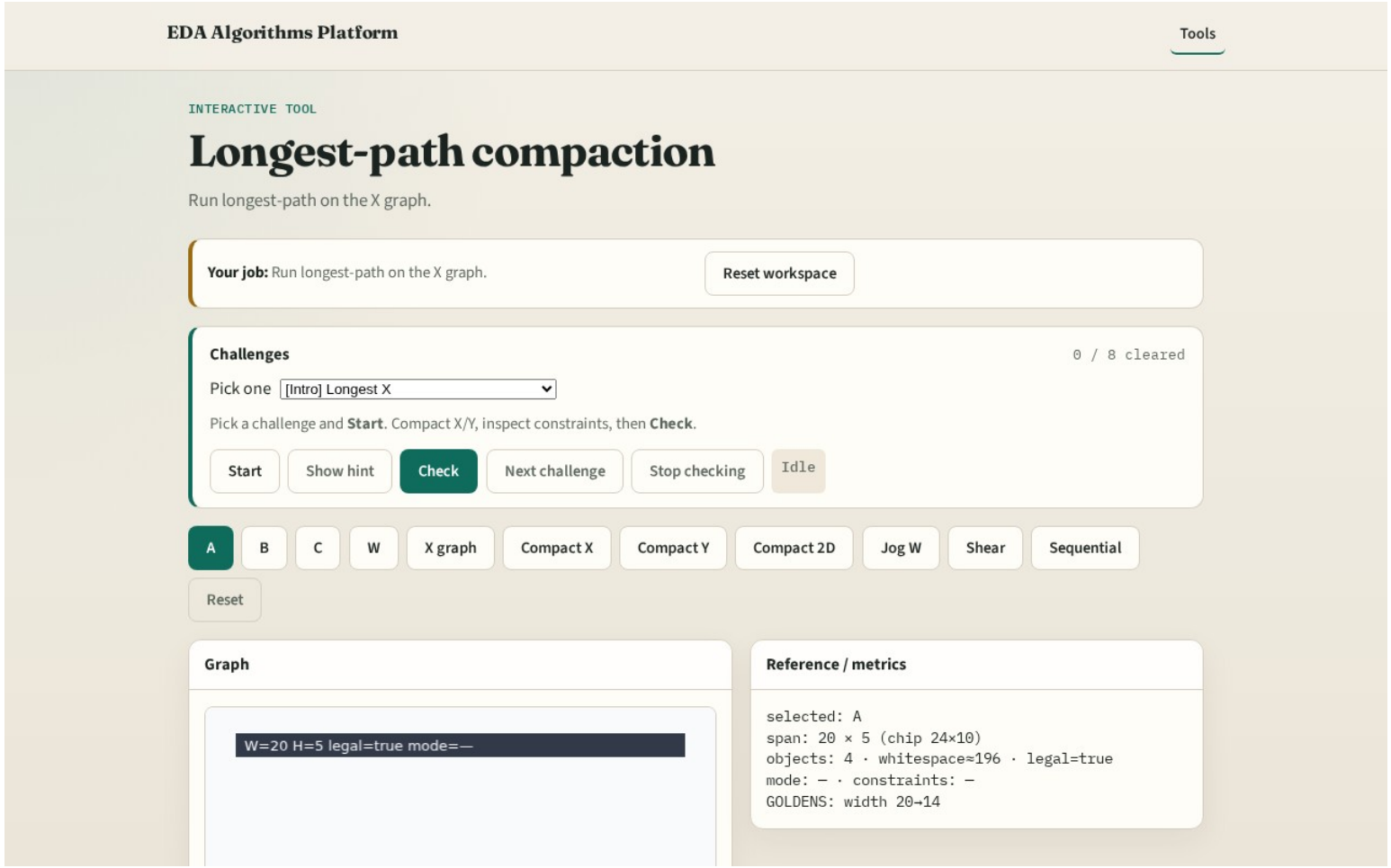
Positions feed compact_1d next.

- apply

```
1-D
```

Ready

Browser lab track



Browser lab starter

Implement track

- Call longest_path_compact(build_constraint_graph(…))

Pitfalls

- Using shortest-path instead
- Not iterating to fixpoint

Your turn

- Next: apply 1-D compaction